# Networks

## Cuts

Michel Bierlaire

Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

# Cuts

*Just as cities are separated into two banks by a river, it may be convenient to separate a directed graph into two sets of nodes.*
*This is called a cut.*

# Cuts

# Directed graph: $(\mathcal{N}, \mathcal{A}, \phi)$

## Cut

A cut $\Gamma$ is

- an ordered partition of the nodes
- into two non empty subsets:

$$\Gamma = (\mathcal{M}, \mathcal{N} \setminus \mathcal{M}),$$

where $\mathcal{M} \subseteq \mathcal{N}$ and $\mathcal{M} \neq \emptyset$.

# Directed graph: $(\mathcal{N}, \mathcal{A}, \phi)$

## Cut

A <u>cut</u> $\Gamma$ is

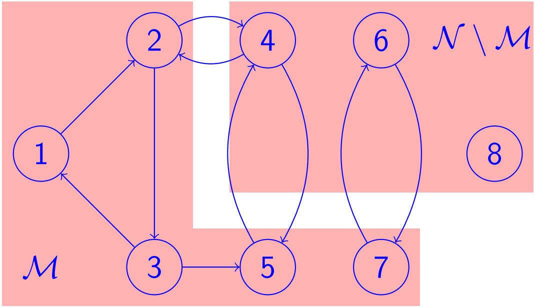
- an ordered partition of the nodes
- into two non empty subsets:

$$\Gamma = (\mathcal{M}, \mathcal{N} \setminus \mathcal{M}),$$

where $\mathcal{M} \subseteq \mathcal{N}$ and $\mathcal{M} \neq \emptyset$.

## Ordered

$$(\mathcal{M}, \mathcal{N} \setminus \mathcal{M}) \neq (\mathcal{N} \setminus \mathcal{M}, \mathcal{M})$$



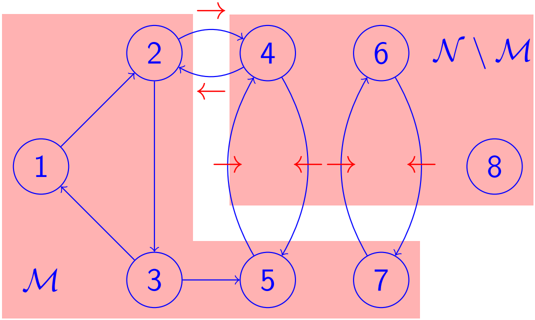
# Definitions

## Forward arcs

$$\Gamma^{\rightarrow} = \{(i,j) \in \mathcal{A} | i \in \mathcal{M}, j \notin \mathcal{M}\}.$$

## Backward arcs

$$\Gamma^{\leftarrow} = \{(i,j) \in \mathcal{A} | i \notin \mathcal{M}, j \in \mathcal{M}\}.$$



# Summary

*Cuts separate the directed graph into two parts.*
*It is useful to focus only on what happens between the two parts*
*Like a river and its bridges.*